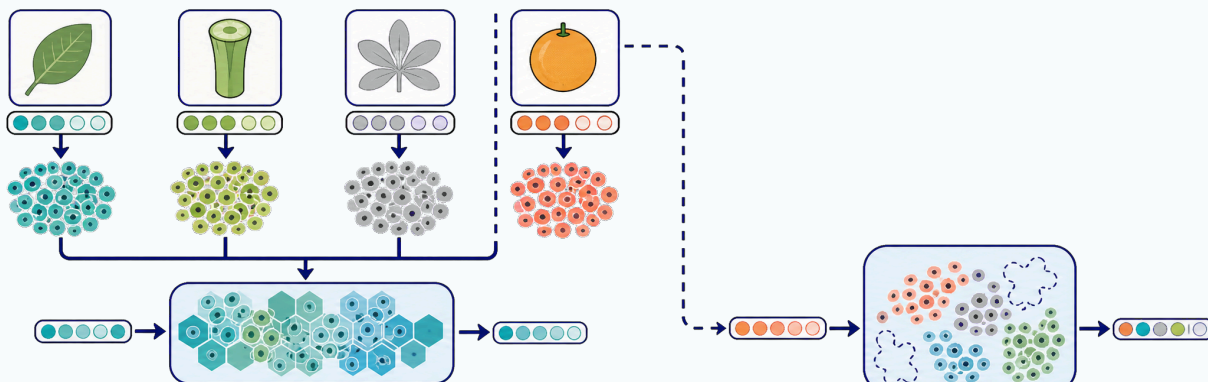


Target-species adaptation landscape from sparse labelled support

8 target species - 10 fixed draws per setting - support/query partitions disjoint

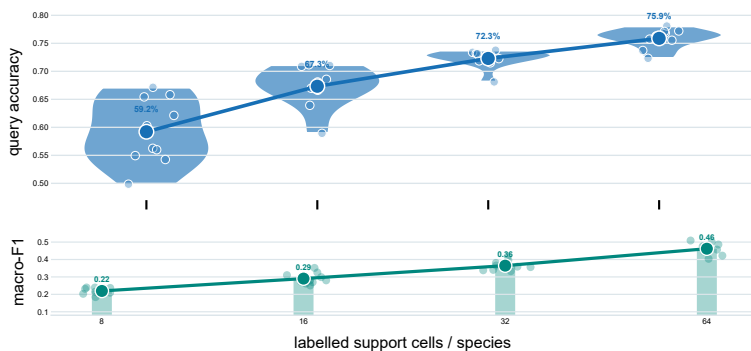
a

Sparse target adaptation landscape

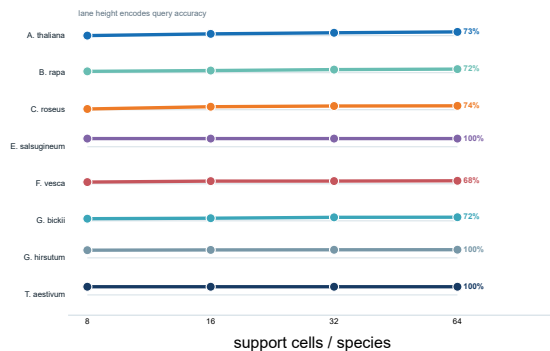


support labels tune low-rank updates; query cells remain disjoint

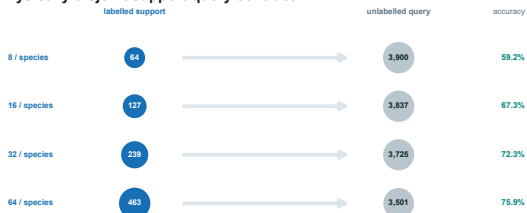
b Support-dose response across independent draws



c Species-resolved adaptation trajectories



d Physically disjoint support/query contract



e Allocation strategy landscape

